

Anomaly Detection by Effectively Leveraging Synthetic Images[‡]

Sungho Kang^{1, †}, Hyunkyu Park^{1, †}, Yeonho Lee¹, Hanbyul Lee², Mijoo Jeong³, YeongHyeon Park¹, Injae Lee¹, and Juneho Yi^{1, *}

¹ Department of Electrical and Computer Engineering, Sungkyunkwan University, Suwon 16419, Republic of Korea; sungho369@skku.edu (S.K.); mjss016@skku.edu (H.P.); tarazed@skku.edu (Y.L.); yeonghyeon@g.skku.edu (YH. P.); injea1291@skku.edu (I.L.); jhyi@skku.edu (J. Y.)

² Department of Artificial Intelligence, Sungkyunkwan University, Suwon 16419, Republic of Korea; byul1748@g.skku.edu (H. L.)

³ Department of Architecture, Sungkyunkwan University, Suwon 16419, Republic of Korea; gor-geousmj2@g.skku.edu (M. J.)

* Correspondence: jhyi@skku.edu

[†] These authors contributed equally to this work.

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Abstract: Anomaly detection plays a vital role in improving productivity in industrial manufacturing. Due to the scarcity of real defect images, unsupervised approaches that rely solely on normal samples have been extensively studied. In contrast, the use of synthetic defect images for training purposes has increasingly gained attention as an alternative solution. However, this alternative approach suffers from high computational complexity and insufficient realism in the generated defect images. To address this problem, we propose a framework that exploits a pre-trained text-guided image-to-image model and an image retrieval model to generate realistic defect images. Specifically, the image retrieval model assesses the similarity of the generated images and filters out irrelevant or meaningless outputs, thereby enhancing the quality and relevance of the synthetic defect images. In addition, we introduce a training strategy that effectively incorporates the synthetic images into anomaly detection models. Experiments on the MVTec AD dataset demonstrate the effectiveness of our approach.

Keywords: anomaly detection; pre-trained text-guided image-to-image translation model; pre-trained image retrieval model; synthetic data; training strategy

1. Introduction

Detecting defective products at an early stage in industrial manufacturing is critical for improving productivity. However, due to the characteristics of manufacturing environments, acquiring real defect images is highly challenging, which makes it difficult to construct paired data for supervised learning. As a result, unsupervised learning methods that solely on normal images have widely been studied.

With the improvement in the quality of synthetic images, generating synthetic defect images for training purposes has emerged as a promising research direction. As illustrated in Figure 1, defect image synthesis methods can be broadly categorized into two types: simple synthesis and generative model-based synthesis. Simple synthesis creates defect images by injecting noise into normal images [1] or by pasting image patches from one area to another [2]. These methods are cost-effective as they do not require a training

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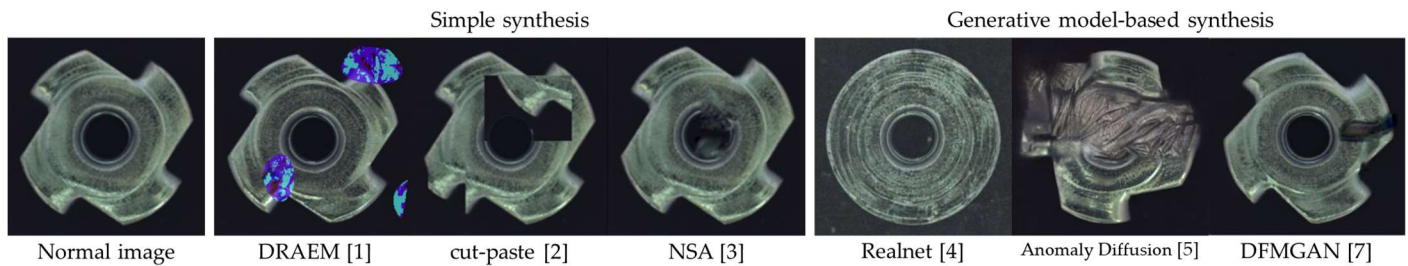


Figure 1. Research on defect image synthesis can be broadly categorized into two approaches: simple synthesis and generative model-based synthesis. Simple synthesis include techniques such as noise-based augmentation [1], patch-based insertion [2], and Poisson-based image blending [3]. On the other hand, generative model-based synthesis utilizes diffusion models [4,5] or GAN [7] to generate realistic defect images.

procedure and allow easy control over defect patterns. However, the resulting synthetic defect images often lack realism and are unlikely to represent real industrial defects. This limitation hinders performance when synthetic defect images are used for training. In contrast, generative model-based synthesis—such as those utilizing diffusion models [4,5,6] or GANs [7,8,9]—can produce highly realistic synthetic defect images. These methods require substantial computational resources for model training and sampling process while offering limited controllability over defect patterns.

Recently, as the quality of synthetic images has improved, increasing attention has been given to training strategies that effectively utilize such data. This strategies can be broadly divided into two categories: strategies that incorporate real images alongside synthetic ones, as well as those that depend solely on synthetic images. Although prior works have proposed numerous strategies, those using synthetic data alone remain underexplored in terms of their effectiveness in anomaly detection performance.

To address these challenges, we propose a novel framework that synthesizes defect images. The framework is composed of a pre-trained text-guided image-to-image translation model and a pre-trained image retrieval model. It generates synthetic defect images by transforming normal images into defect images based on user-defined defect prompts. To verify whether the synthesized defect images are properly generated, the image retrieval model evaluates its similarity with the corresponding normal input images. To enable efficient training using only synthetic data, we introduce a training strategy that combines defect images that are synthesized by a simple synthesis and a generative model-based synthesis. This approach significantly reduces the number of images for training that need to be generated by generative models, thereby lowering the computational cost. We validate the effectiveness of the proposed method through experiments on the MVTec AD dataset [10].

Our contributions are:

- We present a novel training-free defect image synthesis framework that leverages a pre-trained text-guided image-to-image translation model and an image retrieval model to generate realistic defect images.
- We introduce an efficient training strategy that combines simple synthesis and generative model-based synthesis to reduce reliance on computationally expensive generative models.

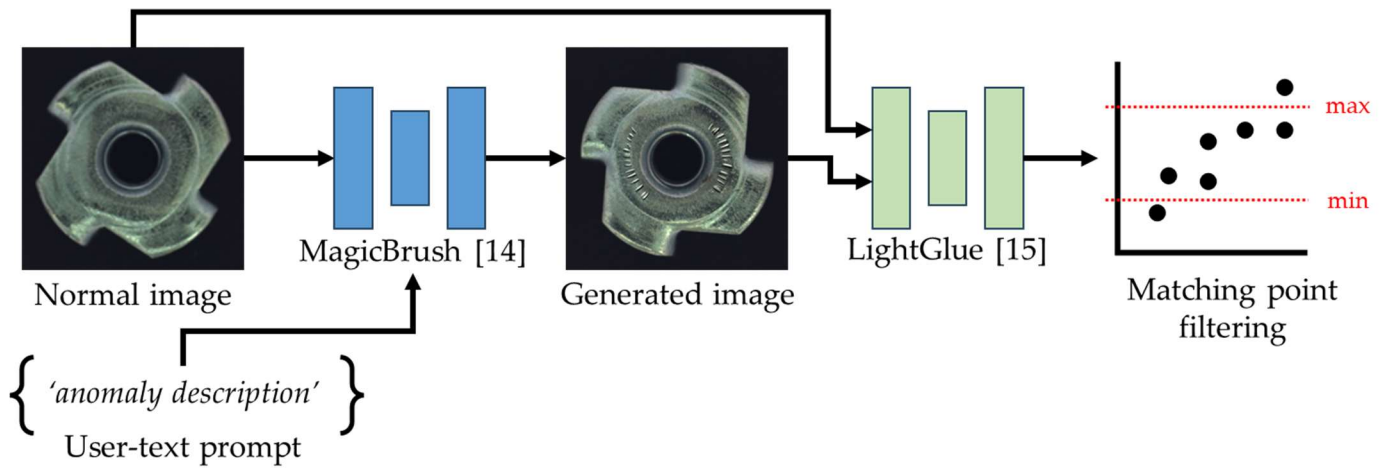


Figure 2. Overview of the proposed framework. It consists of a pre-trained text-guided image-to-image translation model (MagicBrush [14]) and image retrieval model (LightGlue [15]). A normal image is transformed by MagicBrush [14] using a user-text prompt and LightGlue [15] assesses structural similarity between the original and generated image to filter out irrelevant results.

2. Related work

2.1. Synthetic defect image generation

Defect image synthesis has emerged as a critical research direction to address the scarcity of labeled defect samples in industrial anomaly detection. Existing approaches are generally classified into two categories: simple synthesis and generative model-based synthesis. The simple synthesis such as Perlin noise injection [1], cut-and-paste [2], and Poisson blending [3] synthesize defect patterns by directly manipulating normal images without the need for additional model training. These methods are computationally efficient and allow explicit control over the size, shape, and location of anomalies. However, they often fail to capture the visual complexity of real-world defects, limiting their effectiveness.

Generative model-based synthesis that is mostly based on GANs or diffusion models has received huge attention due to its capacity to produce realistic synthetic defect images. Although numerous GAN-based generative methods have been explored [7,8,9], diffusion models have outperformed GANs in image quality and have garnered significant interest in the research community. Realnet [4] introduced a network that generates anomalies with adjustable intensity by injecting noise during the diffusion process, enabling the synthesis of diverse anomaly patterns. Anomaly Diffusion [5] proposed a framework that disentangles appearance and location information to produce diverse anomaly images. DualAnoDiff [6] employed a dual-interrelated diffusion model to simultaneously generate both the anomaly image and its corresponding anomaly part. These methods can produce defect images that better represent real-world defects observed in manufacturing processes, but they require substantial computational cost for training and sampling. To mitigate this issue, we propose a novel framework that leverages a pre-trained text-guided image-to-image translation model and a pre-trained image retrieval model.

2.2. Training strategy using synthetic data

With the improvement in the quality of synthetic images, many studies have explored strategies for utilizing synthetic images as training data. These studies can be broadly categorized into two approaches: those that rely solely on synthetic images [11], and those that leverage both synthetic images and real images [12,13]. Since acquiring real

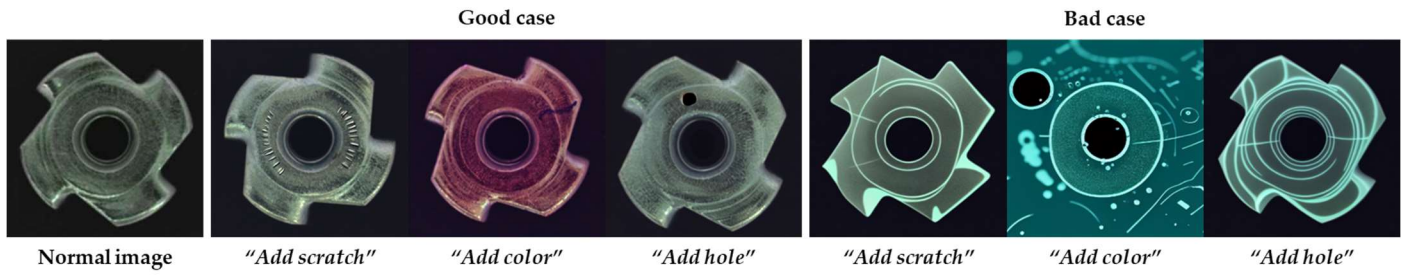


Figure 3. Examples of synthetic defect images that are translated from a normal image. The ‘Good cases’ show structurally consistent synthetic defect images, while the ‘Bad cases’ present inconsistent and irrelevant synthetic defect images. These examples highlight the necessity of filtering out irrelevant results.

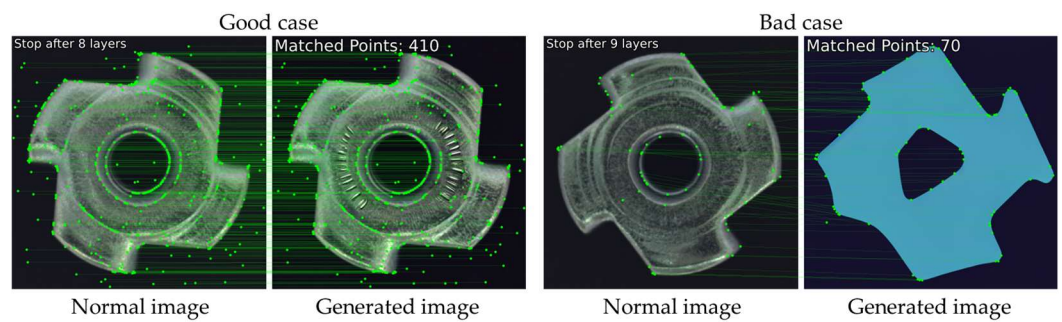


Figure 4. Results of comparing normal and transformed images using LightGlue [15]. For the ‘Good case’, many matching points are matched, indicating structural consistency. In contrast, only a small number of matching points appear, highlighting the structural inconsistency in the ‘Bad case’.

defect images for anomaly detection is extremely challenging, training must rely solely on synthetic data. Moreover, model training requires a large volume of synthetic defect images, which leads to high computational costs. To address this problem, we propose an efficient training strategy that leverages both synthetic defect images created by generative model-based and simple synthesis methods.

3. Methods

3.1. Synthetic Defect Image Generation with Pre-trained Models

As illustrated in Figure 2, our proposed framework consists of a pre-trained text-guided image-to-image translation model and an pre-trained image retrieval model. Numerous text-guided image-to-image translation models have been introduced to modify images according to prompts [14,16]. In contrast to other models [16] that perform global transformations, MagicBrush [14] allows precise editing of specific image regions. Therefore, we employ a pre-trained MagicBrush [14] to generate synthetic defect images by providing prompts that reflect the characteristics of potential abnormal patterns.

However, since the pre-trained MagicBrush [14] is trained on the MS COCO dataset [17], it may generate results that are unrelated to the input image, as shown in Figure 3. Training of the anomaly detection model with unrelated results such as the bad cases in Figure 3 can severely degrade its performance. To prevent such cases, we utilize LightGlue [15], a pre-trained image retrieval model. LightGlue [15] matches local features between images and evaluates structural similarity based on the matched points, as shown in Figure 4. By adjusting the range of the number of matching points, LightGlue [15] can filter out irrelevant images such as those with excessive changes or no meaningful changes.

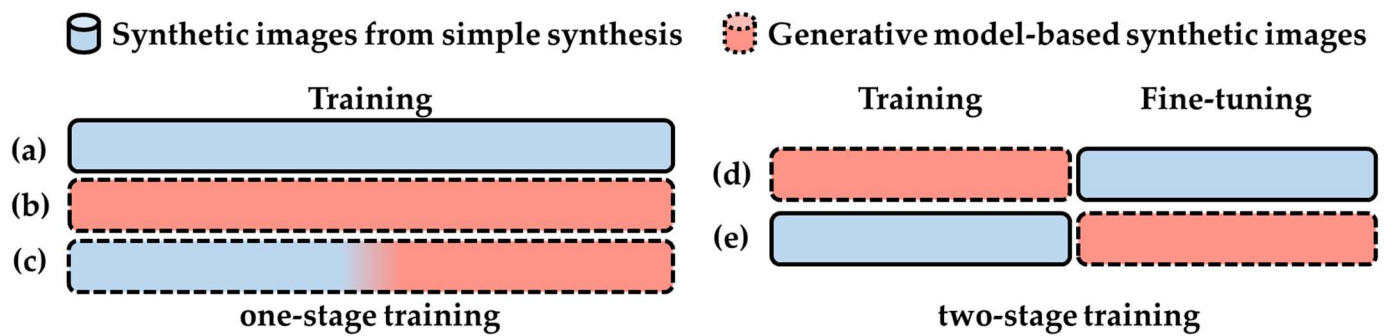


Figure 5. Overview of the training pipeline. Blue boxes represent defect images generated by simple synthesis, while red boxes represent generative model-based synthetic images. Strategies (a), (b), and (c) use only one-stage training, whereas strategies (d) and (e) employ a two-stage training process with fine-tuning.

Table 1. Comparison of anomaly detection performance across training strategies. The proposed approach, (e), consistently outperforms other strategies.

Strategy class	(a)	(b)	(c)	(d)	(e)
capsule	97.3	87.1	97.4	54.2	98.7
bottle	96.7	94.8	96.3	96.1	99.0
pill	98.2	81.8	97.0	76.2	98.4
transistor	93.1	89.8	92.1	83.1	95.9
cable	91.8	70.4	92.2	80.9	97.1
toothbrush	99.7	99.4	99.4	77.8	100
metal nut	98.7	84.4	97.5	64.1	99.1
hazelnut	100	88.0	100	94.3	100
screw	93.9	89.5	96.2	70.3	99.5
avg	96.6	87.2	96.5	77.4	98.7

3.2. Training strategy

Training of an anomaly detection model solely with synthetic images requires a large amount of sampling, which in turn demands substantial computational resources. To address this issue, we propose a two-stage training strategy that leverages both simple synthesis and generative model-based synthesis as depicted (e) in Figure 5. During the first stage, simple synthesis is used to efficiently generate a large number of synthetic defect images for pre-training the model. In the second stage, a smaller set of synthetic defect images generated by a generative model-based synthesis is leveraged to fine-tune the model. This training strategy requires minimal computational resources while achieving a reasonable performance.

4. Experiments

For experiments, we have employed the publicly available anomaly detection dataset MVTec AD [10]. This dataset consists of nine object categories and six texture categories.

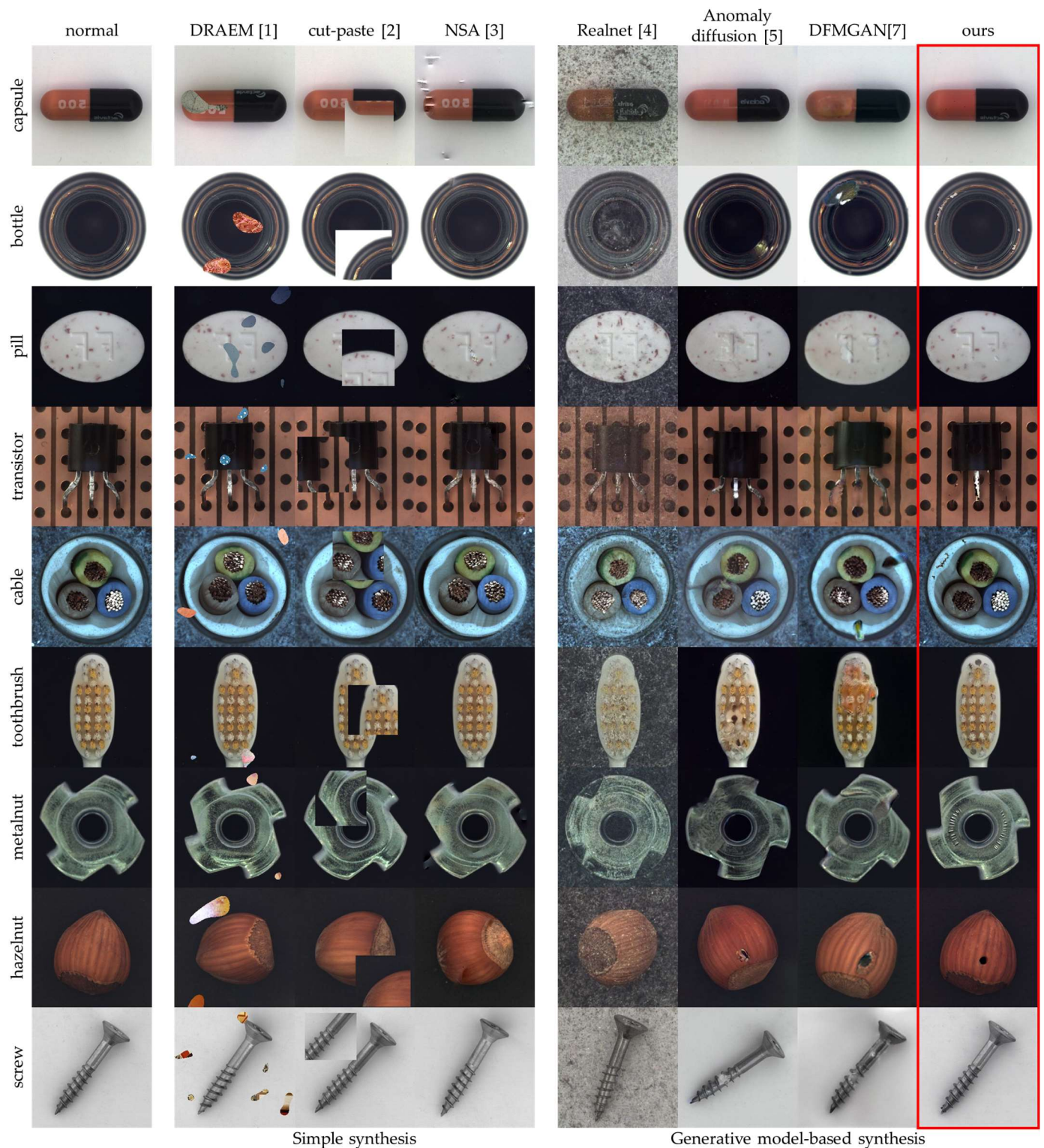


Figure 6. Qualitative comparison of defect image synthesis methods across various object categories from the MVTec AD dataset [10]. From left to right: Dream [1], Cut-Paste [2], NSA [3], RealNet [4], AnomalyDiffusion [5], DFM-GAN [7], and Ours. Defect images synthesized using simple synthesis methods [1,2,3] are often unrealistic and do not resemble defects that can occur in real-world industrial manufacturing environments. In contrast, generative model-based synthesis approaches [4,5,7] generally produce more realistic defect images. Our proposed method, despite relying solely on pre-trained models, is capable of generating defect images that are even more realistic than those produced by existing generative synthesis methods.

Among these, we have only used nine object categories such as capsule, bottle, pill, transistor, cable, toothbrush, metalnut, hazelnut, and screw for our experiments. For generative model-based synthesis, three hundred synthetic defect images are generated using MagicBrush [14] and LightGlue [15], with pre-trained models that are available from their official GitHub repositories. For simple synthesis, we adopted the method used in DRAEM [1]. To validate the effectiveness of synthetic defect images and the proposed training strategy, we have conducted experiments for DRAEM [1].

To ensure a fair evaluation of the generated images, we compare our method with both simple synthesis methods—Dream [1], Cut-Paste [2], NSA [3]—and generative model-based synthesis methods—RealNet [4], AnomalyDiffusion [5], DFM-GAN [7] as illustrated in Figure 6. Our proposed framework not only surpasses the simple methods in terms of realism and quality but also achieves comparable performance to generative models, despite employing only pre-trained models.

To confirm the effect of training strategies, we have conducted experiments using various configurations, as illustrated in Figure 5. Training strategies (a), (b), and (c) involve a single stage training process with different types of datasets, whereas strategies (d) and (e) perform an additional fine-tuning stage after the initial training. As shown in Table 1, models trained on simple synthetic images achieved the highest performance in single-stage training. In contrast, the dataset generated by our proposed method, which contains a relatively small number of samples, led to the lowest performance. However, when the model is initially trained on simple synthetic images and subsequently fine-tuned with the images generated by the proposed method, it achieves the best performance. These results suggest that pretraining with a large amount of synthetic images helps establish a strong baseline performance, and further fine-tuning with generative model-based synthetic images enables the model to better adapt to realistic anomalies, significantly improving its detection performance.

5. Conclusions

Recent advances in synthetic image generation have led to growing interest in utilizing synthetic images as training data for anomaly detection. However, their high computational cost poses a challenge for large-scale deployment. We propose a framework that utilizes pre-trained models to synthesize realistic synthetic defect images. In addition, we introduce an effective training strategy that leverages both simple synthesis and generative model-based synthesis. Our experimental results confirm the effectiveness and practicality of the proposed framework and training strategy. In future work, we plan to further explore synthesis strategies tailored to different object types and defect categories and investigate methods for controlling the size and location of defects when generating high-quality anomaly images without additional model training.

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